

Retrieval-based forecast adaptation

Executive summary of results

Adaptation pipeline and notations

For query lookback X and target $T \in \mathbb{R}^H$, the frozen backbone F produces the vanilla forecast $V = F(X)$. Retrieval returns K historical pairs (X_k, Y_k) with distance weights w_k ; their backbone forecasts are $N_k = F(X_k)$, their weighted future is $\bar{Y} = \sum_{k=1}^K w_k Y_k$, and the retrieved pairs also produce the covariate-conditioned forecast $C = F(X; \{(X_k, Y_k)\}_{k=1}^K)$. The adaptor maps $(V, C, Y_{1:K}, N_{1:K}, \bar{Y})$ to the final forecast $\hat{T}$. For a candidate adaptation A , its realized advantage over vanilla is

$$\Delta(A) = \frac{1}{H} \sum_{h=1}^H \left[(T_h - V_h)^2 - (T_h - A_h)^2 \right].$$

We propose various *baselines* corresponding to different formulas fitted on $C = F(X; \{(X_k, Y_k)\}_{k=1}^K)$ to predict T , as well as *gates* which take binary decisions to choose between V and a given non-trainable adaptor $A \in \{C, \bar{Y}\}$.

Practical details

Component	Protocol
Datasets	Electricity, Traffic, Solar, and <code>exchange_rate</code> .
Settings	$(L, H) \in \{(168, 24), (336, 48), (504, 168)\}$: 12 dataset–setting configurations.
Backbone	Frozen <code>chronos2</code> .
Splits	T_0 datastore (30%); T_1 training (40%); T_2 validation (20%), T_3 eval (20%).

For a given configuration c and method m , relative improvement in nMSE over the vanilla run from the *same retrieval pipeline* is

$$I_{m,c} = 100 \frac{\text{nMSE}_{v,c} - \text{nMSE}_{m,c}}{\text{nMSE}_{v,c}} \%.$$

Tables report the mean of $I_{m,c}$ over the 12 held-out T_3 configurations.

Selected baseline and gate candidates

Code name	Family / design	Space	K	T3 gain
<code>full_ridge_shared</code>	ridge on $V, C, Y_{1:K}, N_{1:K}$	instance	3	+1.951%
<code>full_ridge_shared</code>	ridge on $V, C, Y_{1:K}, N_{1:K}$	raw	1	+1.789%
<code>cov_y_ridge_shared</code>	ridge on $V, C, Y_{1:K}$	instance	3	+1.782%
<code>full_ridge_shared</code>	ridge on $V, C, Y_{1:K}, N_{1:K}$	raw	3	+1.780%
<code>cov_avgy_ridge_shared</code>	ridge on $V, C, \bar{Y}$	instance	3	+1.765%
<code>catboost_cov_regressor_shared</code>	$G(C; g_\theta(z))_h$	raw	1	+0.031%
<code>catboost_avgy_regressor_shared</code>	$G(\bar{Y}; g_\theta(z))_h$	instance	1	-1.900%
<code>bayes_cov_shared</code>	$G(C; b(C))_h$	raw	1	0.000
<code>bayes_avgy_shared</code>	$G(\bar{Y}; b(\bar{Y}))_h$	instance	1	0.000

The corrected ranking carries each result’s retrieval identity to its matching vanilla reference. The five manually selected baseline pipelines improve mean held-out nMSE; `full_ridge_shared` with instance retrieval and $K = 3$ is best at +1.951%. The selected gates do not provide a comparable gain.

Completed current ablations

All comparisons below use the corrected retrieval-matched vanilla reference.

Axis	Equal-configuration result	Conclusion
Fourier retrieval	Selected ridges gain 1.76–1.95% with their original retrieval and 1.28–1.62% with FFT amplitude.	Do not promote Fourier.
Fixed T_0 store	Every paired ridge is at least as good with fixed retrieval: 1.80–2.09% versus 1.79–1.95% online.	Prefer the stricter fixed store.
Per-horizon ridge	Shared controls gain 1.76–1.95%; horizon fits gain 1.33–1.52%.	Extra horizon parameters hurt.
Convex fit	Matched convex variants range from -0.05% to $+0.88\%$, below every shared-ridge control.	Reject convex variants.
Δ -ridge fit	Matched variants gain 0.47–0.82%, again below every shared-ridge control.	Reject Δ -ridge variants.
Lookback L at $H = 24$	$L = 168$ gives the most balanced full-ridge gains (0.81–1.14%); $L = 504$ gives 0.24–0.41%. The $L = 24$ mean is distorted by a Traffic nMSE of 2.07×10^{10} .	Keep $L = 168$ as the stable choice.
Mixed-quantity panels	Weather worsens for every selected ridge; ETTm1 raw retrieval and the ETTm2 avgY gate show catastrophic outliers.	Keep these panels ablation-only.

Baseline lookback-length ablation (equal mean over four datasets)

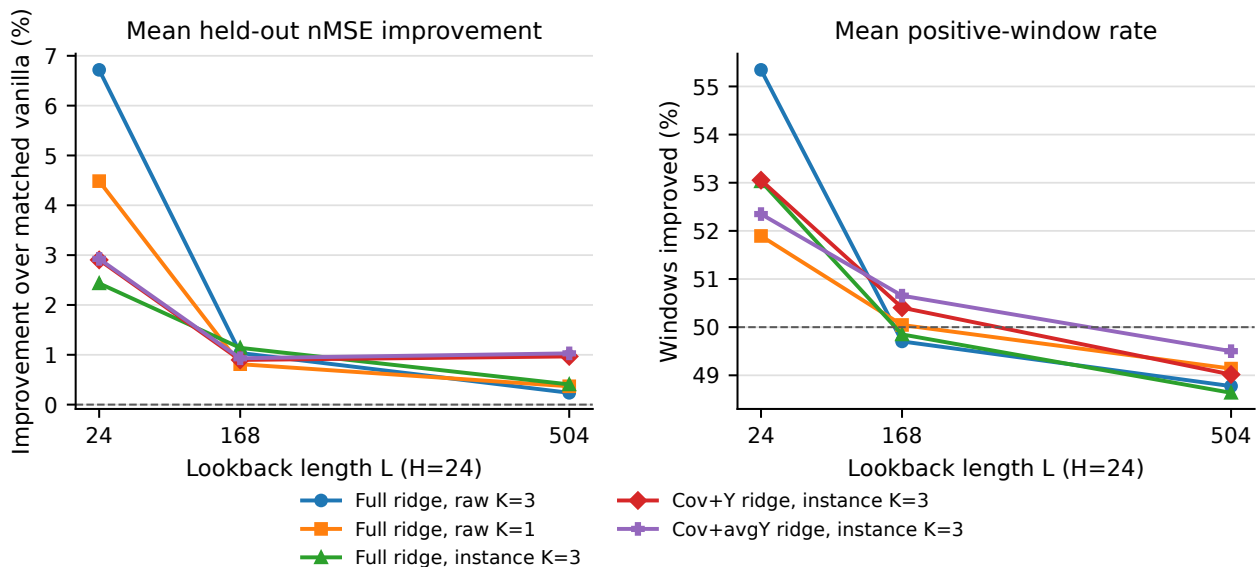


Figure 1: Completed baseline lookback ablation at fixed $H = 24$. $L = 24$ has the largest numerical means but includes the pathological Traffic scale; $L = 168$ is the balanced retained setting.

Partial and failed workflows

TS-RAG (job 43579). All four released-model inferences completed, but the table step failed to discover them. Against matching Chronos-Bolt vanilla, TS-RAG changes nMSE by $+0.44\%$ (Electricity), $+0.48\%$ (Traffic), -2.01% (Solar), and $+8.69\%$ (Exchange): $+1.90\%$ equally averaged. Against the selected full-ridge control the mean advantage is only $+0.85\%$, with losses on Traffic and Solar. The metric artifacts are valid; the workflow/table is not terminally successful.

Exact-split SOTA benchmark (job 43580). Five of seven local rows completed before Electricity retrieval was OOM-killed; Exchange did not start. The adapted MSE is 0.3446/0.2500/0.3077/0.1494/0.2079 on ETTh1/ETTh2/ETTh1/ETTh2/Weather. It improves local vanilla on four rows, but its five-row mean (0.2519) remains above the published TS-RAG/Cross-RAG means (0.2398/0.2322). This is not yet a complete benchmark.

K and horizon sweeps. Job 43589 stopped on the now-removed $K = 100$. The active $K \leq 20$ grid has 60/144 current retrieval extractions complete; a rerun must finish the remaining 84 before adaptation and tables. Job 43592 completed the raw- $K = 1$ gate branch, then collided with the identical Electricity 504:24 instance- $K = 3$ extraction concurrently owned by the L job. That extraction is now complete and reusable, but H still needs a rerun.

TS-IFA results

Code name	Space / steps	T2 gain	T3 gain	Checkpoint / held-out behaviour
joint_ridge_horizon_unconstrained_full	raw / 10K	+1.84%	+0.04%	Window gains +3.26, +0.58, −3.72%; all three select 10K.
joint_ridge_horizon_unconstrained_full	raw / 30K	+3.03%	−6.71%	Selected at 29/20/20K for (168/24, 336/48, 504/168).
joint_ridge_horizon_unconstrained_full	instance / 30K	+3.25%	−17.97%	Selected at 30/26/20K for the same windows.
joint_neural_horizon_unconstrained_full	raw / instance, 30K	−	−4.60/ −2.14%	Zero T3 wins.

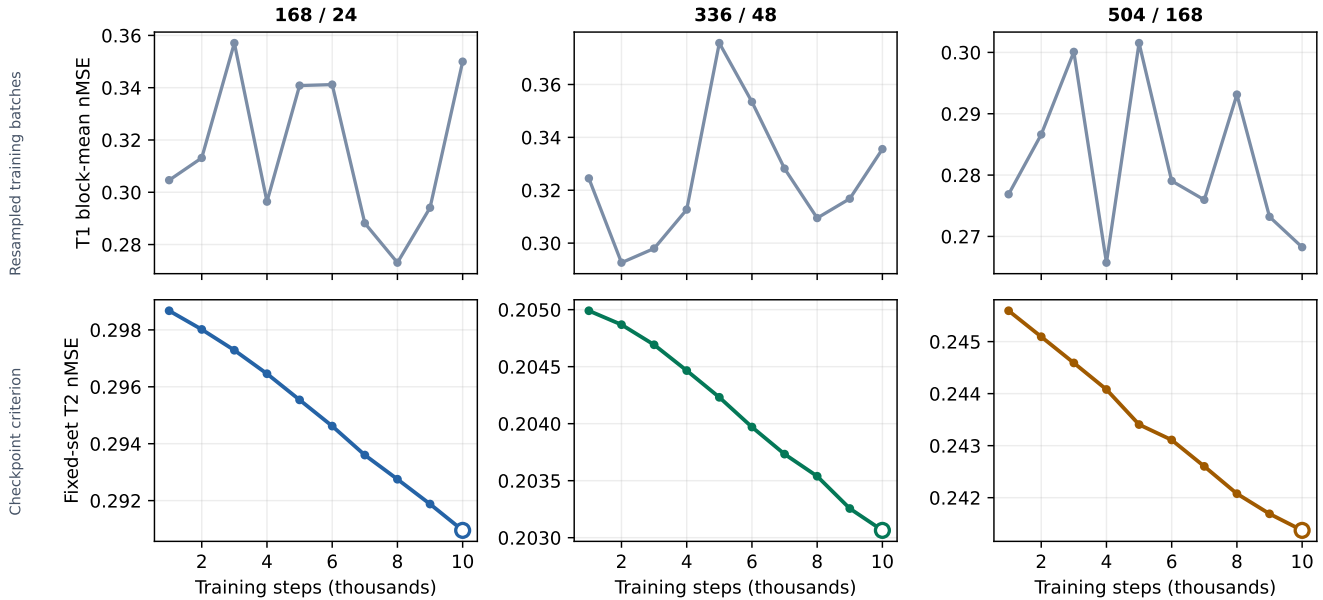


Figure 2: The 10K joint-ridge criterion curves. The gray T1 trace is a 1,000-step block mean over freshly resampled training batches, not a repeated evaluation on one fixed set, so it need not be monotone. The colored T2 trace uses a fixed set and selects 10K for all three windows.

TS-IFA is not reliable on held-out T3. Longer runs improve T2 selection but worsen T3, with the largest failures at $H = 168$; the raw 10K ridge pilot remains the least-bad aggregate result at +0.04%.